AIoT Smart Home Simulation

Public Portfolio Report

Modular ESP32-S3 Edge-AIoT smart home system for comfort, privacy, security and environmental safety

This public version preserves the main technical report and references while removing academic submission material, personal identifiers, private appendix links, user-guide appendices and dashboard screenshot pages that are not suitable for a public repository.

Technologies: ESP32-S3 · Wokwi · Arduino/C++ · Blynk · Edge Impulse · IoT · TinyML

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Sanitised for GitHub/portfolio publication

Table of Contents

Public portfolio report - PDF page numbering includes the cover and this contents page.

1 Introduction 3

2 Literature Review 3

3 System Design 4

3.1 Overall House Architecture 4

3.2 Common Control Philosophy 5

3.3 Away Mode Security Logic 5

3.4 Bedroom Module 6

3.5 Kitchen / Living Room Module 13

4 AI Model Design & Deployment 18

4.3 Dataset Collection and Labelling 19

4.5 Training and Performance Metrics 20

5 Implementation 22

5.2 Bedroom Module Implementation 23

5.3 Kitchen/Living Room Implementation 23

5.4 Blynk Remote Monitoring Layer 25

5.5 Authentication and Security Logic 26

6 Evaluation 26

7 Ethical, Privacy & Security Considerations 27

8 Discussion 27

9 Conclusion & Future Work 28

References 29

Sanitised public version: academic cover sheets, personal identifiers, private links, dashboard screenshots and detailed virtual-pin tables have been removed.

1 Introduction

Artificial Intelligence of Things (AIoT) is an important area within embedded systems, smart environments, and edge computing. Rather than only connecting sensors and actuators, AIoT combines sensing, local reasoning, and real-time response to create context-aware behaviour (Al-Fuqaha et al., 2015; Shi et al., 2016). Smart homes are one of its clearest applications because they bring together comfort, privacy, safety, energy management, and user-focused automation (Stojkoska and Trivodaliev, 2017; Ray, 2018).

This project develops a modular smart home prototype using ESP32-S3 room controllers. Each room works as an independent embedded module with its own sensors, actuators, local logic, and remote monitoring interface, while still forming part of a wider house architecture. This room-level approach improves modularity, explainability, and scalability without relying on a single central controller.

The prototype focuses on a bedroom and a kitchen/living room. Both follow the same overall design philosophy but serve different priorities. The bedroom emphasises comfort, privacy, rest, and support for vulnerable users, whereas the kitchen/living room prioritises functional control, environmental monitoring, gas safety, and emergency response. This reflects a core principle of the project: smart-home behaviour should adapt to the role of each domestic space rather than apply identical logic everywhere.

Technically, the project combines low-cost embedded sensing, explainable rule-based control, Blynk-based IoT interaction, and an Edge Impulse workflow. The bedroom remains mainly rule-based because its behaviour can be expressed through clear comfort and privacy rules. The kitchen/living room extends this with an environmental safety classifier trained in Edge Impulse using gas score, temperature, and humidity. The model predicts five states: `NORMAL`, `COOKING_FUMES`, `GAS_DANGER`, `HEAT_ANOMALY`, and `GAS_HEAT_EMERGENCY`.

The main aim is to design and prototype a modular AIoT smart home that supports local decisions for comfort, privacy, safety, and security. To achieve this, the project develops room-specific embedded logic, integrates sensors and actuators, evaluates behaviour through simulation, and demonstrates an initial embedded machine-learning workflow for environmental safety classification in the kitchen/living room.

2 Literature Review

2.1 AI in IoT Systems

AIoT extends the Internet of Things by moving part of the decision process closer to the device. In conventional IoT systems, sensors often send data elsewhere for processing; in AIoT, more reasoning can happen locally. In domestic environments, this is valuable because latency, connectivity dependence, and privacy concerns can reduce the usefulness of a smart system. The literature shows applications in occupancy detection, environmental control, anomaly monitoring, adaptive lighting, and assisted living. However, the degree of intelligence varies widely, from threshold-based automation to full cloud or edge machine-learning pipelines (Al-Fuqaha et al., 2015; Ray, 2018).

2.2 Edge AI and Local Reasoning

A key distinction is between cloud-centred and edge-centred intelligence. Cloud-based AI supports stronger remote computation and easier large-scale management, but can introduce latency, connectivity dependence, and privacy concerns. In domestic systems, responses to motion, comfort, privacy, or gas-related events are often most useful when they occur locally and immediately. For this reason, edge AI has become increasingly important in smart environments. Even when a full learned model is not deployed, the same principle still applies: room-level local reasoning can make the system faster, more resilient, and easier to trust (Shi et al., 2016).

Recent reviews of edge AI and edge machine learning reinforce this design choice. Edge AI is increasingly used to move inference towards resource-constrained devices, reducing latency and cloud dependence while improving privacy and resilience (Merenda et al., 2020; Singh and Gill, 2023). In the IoT context, TinyML is also presented as a practical way to run lightweight models close to the sensor source, which matches the ESP32-S3 and Edge Impulse direction of this project (Dutta and Bharali, 2021).

2.3 Smart Home Automation and Environmental Control

Smart home literature often treats lighting, thermal control, assisted care, and security as separate functions. Lighting is commonly based on occupancy or brightness thresholds, thermal control on temperature and humidity ranges, and security on simple intrusion alarms. These approaches are useful but can become limited when context is ignored. A stronger design should allow the same input to produce different responses depending on room purpose, time, and user profile. This supports the modular room-specific structure used in this project (Stojkoska and Trivodaliev, 2017).

2.4 Gas and Domestic Safety Monitoring

Gas and smoke sensing is especially relevant in kitchen environments. However, low-cost gas sensors do not always separate harmless vapour, cooking fumes, and dangerous gas-related events clearly. The practical design challenge is therefore to remain sensitive to risk without becoming too reactive during everyday use. A common response is to combine thresholds with persistence over time, contextual interpretation, and secondary environmental signals. This directly informs the gas-safety logic used in the kitchen/living room module.

2.5 Machine Learning for Embedded Safety Classification

TinyML and embedded machine learning are useful when a system must classify patterns from several sensor values without sending data to a remote service. However, safety-related machine learning must be introduced carefully. A learned model should not replace deterministic protection logic unless it has been tested thoroughly in the target environment. For this reason, the project adopts a hybrid approach in which Edge Impulse provides environmental safety classification while rule-based gas escalation remains available as a deterministic fallback (Warden and Situnayake, 2019). This also explains why the project uses a small tabular classifier rather than a heavier model. In embedded IoT systems, model choice must consider memory, latency, energy use, and deployment reliability, not only predictive accuracy (Merenda et al., 2020; Dutta and Bharali, 2021).

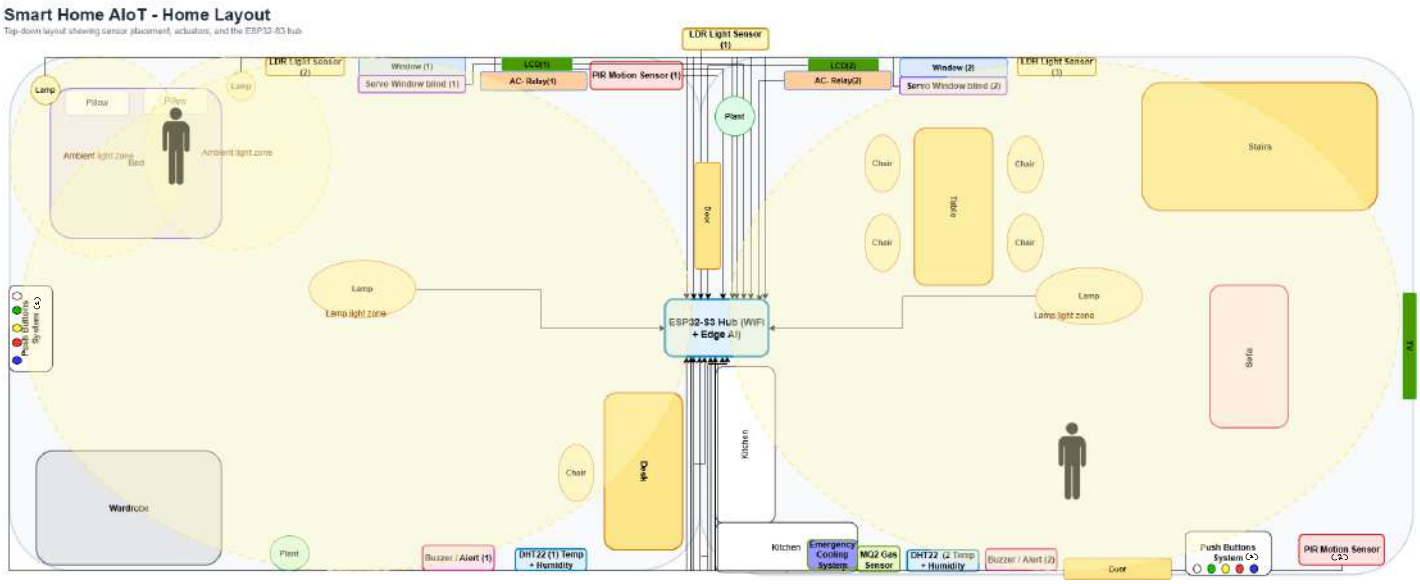
2.6 Research Gap

Many smart-home projects remain either too generic or too centralised. They often apply similar logic across all rooms, rely heavily on cloud services, or prioritise novelty over explainability. There is therefore value in a modular edge-oriented architecture where each room keeps its own priorities while still contributing to a coherent house-wide design. This project addresses that gap through two differentiated room modules built around a shared control philosophy.

3 System Design

3.1 Overall House Architecture

The proposed smart home uses a modular architecture in which each room is represented by an ESP32-S3 controller. Each controller gathers room-specific sensor data, interprets the local context, and applies local decision logic without depending on a permanent cloud service. The current prototype includes a bedroom module and a kitchen/living room module. They are implemented as separate subsystems but share common ideas: operating modes, user profiles, occupancy awareness, local alerts, and safety overrides. This creates house-level consistency while still allowing each room to behave differently according to its function.





(a) 2D plan view.

(b) 3D plan view.

Figure 2: House visualisation created in Planner 5D, showing the 2D plan and the 3D view. Available at: Planner 5D.

3.2 Common Control Philosophy

The house follows a shared logic based on three common room modes (*Normal*, *Privacy*, and *Eco*), three user profiles (*Standard*, *Elderly*, and *Child*), and a wider *Home/Away* house state. *Sleep* is kept only for the bedroom because it is primarily relevant to rest-oriented behaviour. This framework gives the system consistency while still allowing room-specific responses.

3.3 Away Mode Security Logic

Unlike the comfort-oriented room modes, *Away* is treated as a house security state. When the system is armed, comfort behaviour is reduced and the priority becomes monitoring, intrusion escalation, authenticated recovery, and energy saving. Motion is evaluated within a security window so that the system can distinguish between passive monitoring, suspicious motion, and confirmed intruder alert. The design also includes pet-aware behaviour to reduce false alarms caused by small domestic movement.



Figure 3: Away mode security logic across the implemented smart room modules. As shown in this figure, Away mode works as a security layer above the normal room modes. In this state, blinds remain in a defensive position, ordinary comfort functions are reduced, and repeated motion can escalate from monitoring to suspicious motion and then to intruder alert. After authentication, the system returns to Home state instead of staying armed, which helps avoid a repeated alarm loop.

3.4 Bedroom Module

3.4.1 Design Rationale

The bedroom is designed around comfort, privacy, rest, and support for vulnerable users. It therefore includes adaptive blinds, ambient support lighting, a manual lamp state, thermal monitoring, AC output, and occupancy-aware logic. Compared with the kitchen/living room, it requires finer behavioural distinctions, especially at night and across user profiles.

3.4.2 Hardware Structure

The bedroom module includes an ESP32-S3, PIR motion sensor, indoor and outdoor LDR sensors, a servo-controlled blind, smart ambient light output, manual lamp state indicator, DHT22 temperature and humidity sensor (Aosong Electronics, n.d.), AC relay, buzzer, control buttons, and an LCD interface.

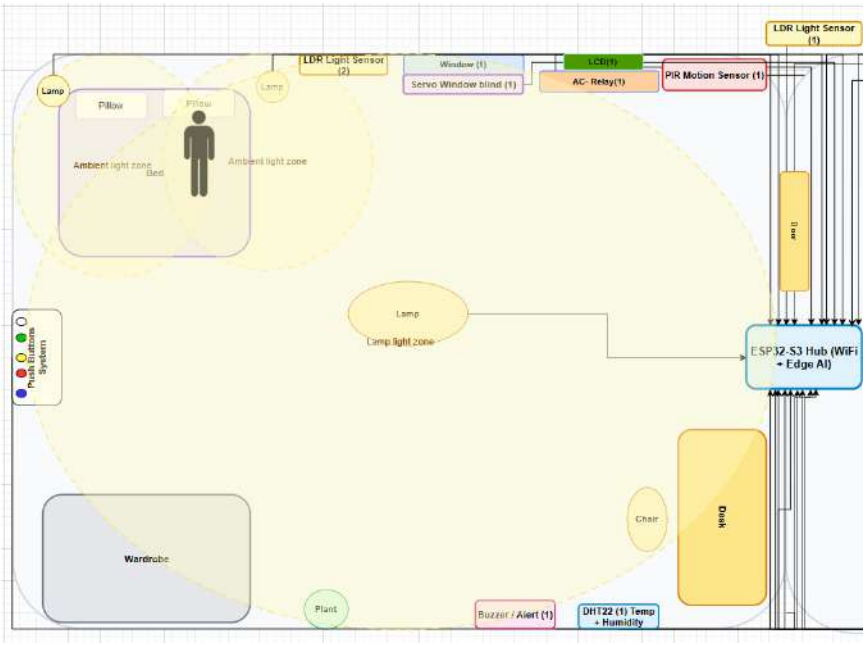


Figure 4: Bedroom hardware layout and logical arrangement in draw.io.



(a) 2D plan view.

(b) 3D view 1.

(c) 3D view 2.

(d) 3D view 3.

Figure 5: Bedroom visualisation created in Planner 5D, including the 2D plan and three dimensional views. Available at: Planner 5D.

3.4.3 Bedroom Behavioural Logic

The bedroom uses four modes. *Normal* supports balanced everyday behaviour, *Sleep* prioritises rest, *Privacy* restricts visibility and daylight exposure, and *Eco* reduces unnecessary energy use while keeping basic support available. Profiles refine this behaviour further: *Standard* provides the default response, *Elderly* extends presence hold time and favours gentler night support, and *Child* adds similarly protective behaviour together with protected deactivation through PIN logic.

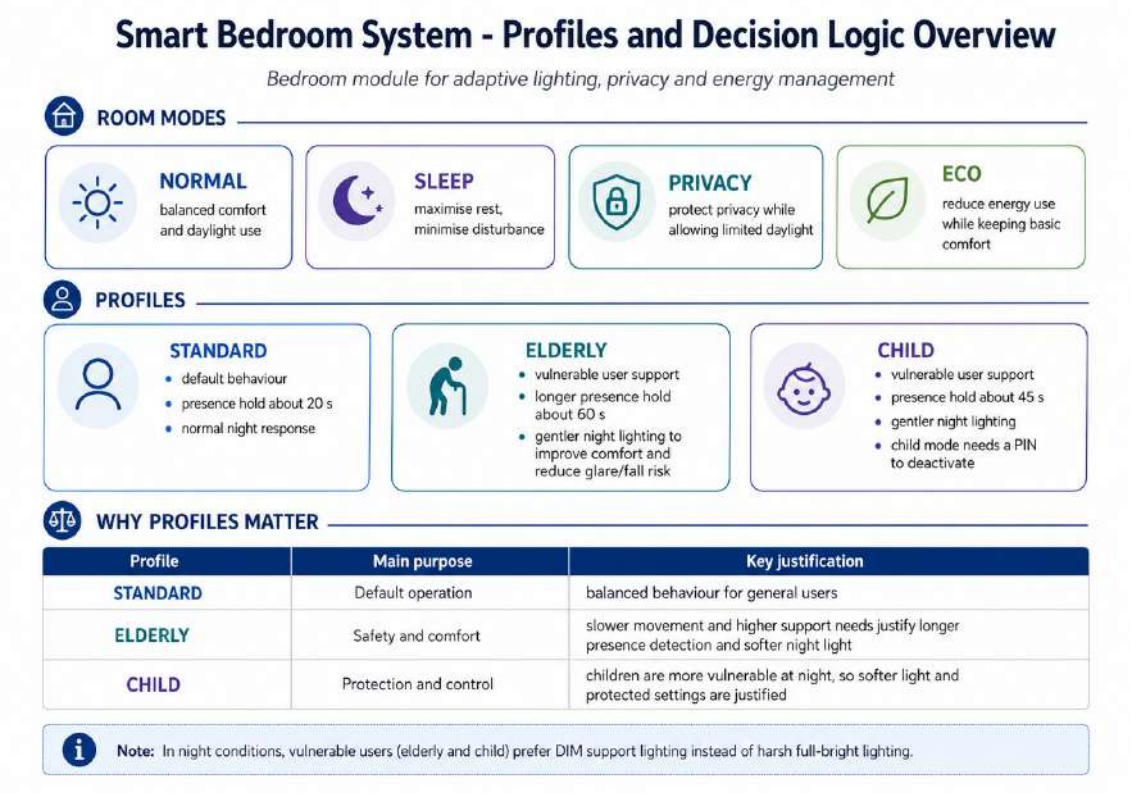
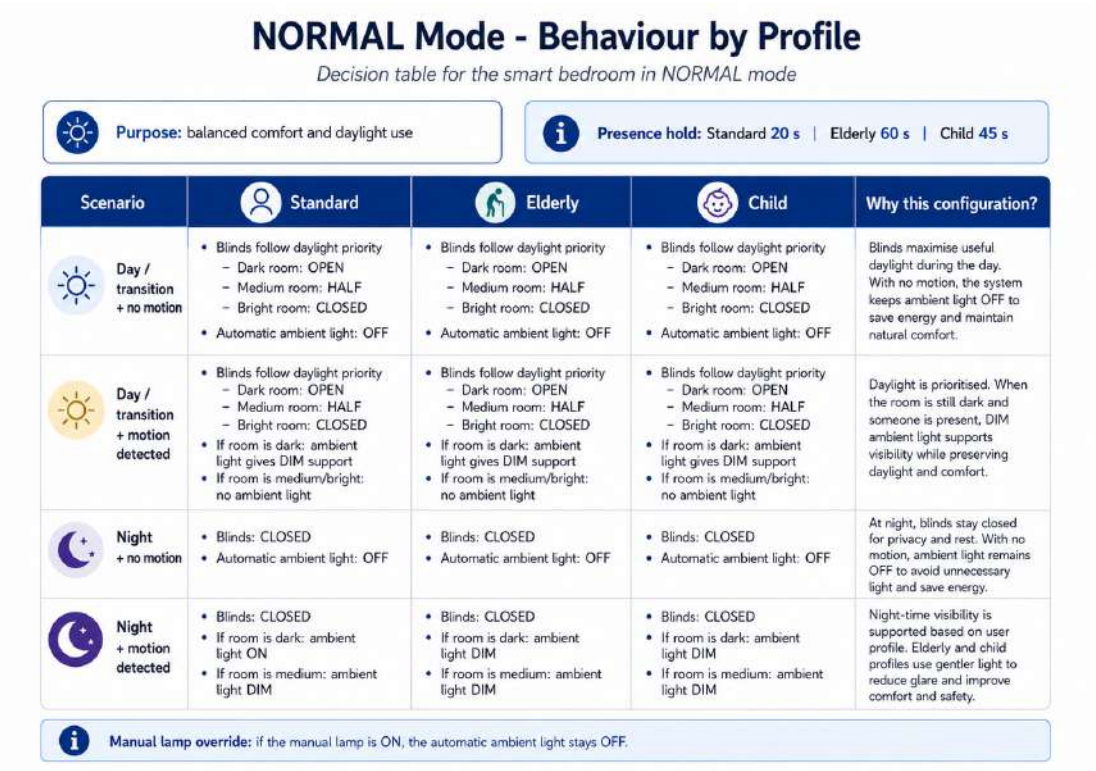


Figure 6: Bedroom profiles and overall control logic overview.

3.4.4 Bedroom Lighting and Blinds Logic

Lighting and blind behaviour are central to the bedroom design. *Sleep* mode keeps the room as dark as possible, while the *Elderly* and *Child* profiles still allow limited night support in selected motion situations. The detailed decision tables below summarise the designed responses.



NORMAL Mode - Behaviour by Profile

Decision table for the smart bedroom in NORMAL mode

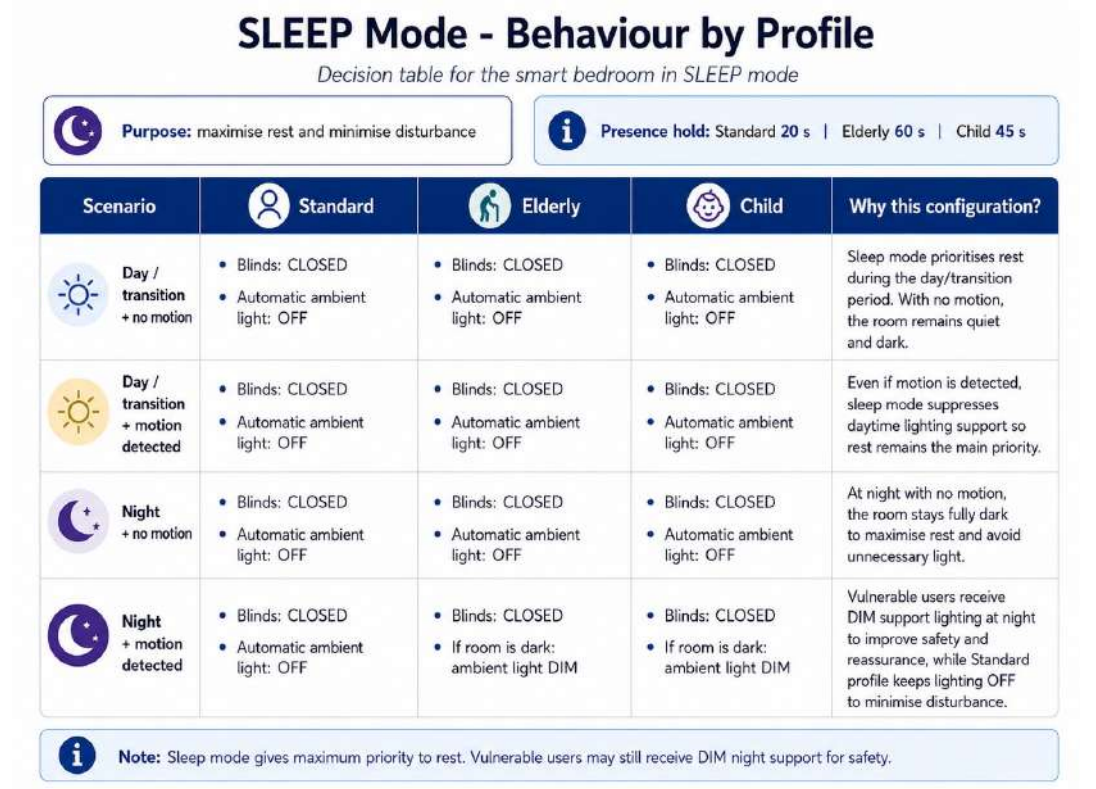
Purpose: balanced comfort and daylight use

Presence hold: Standard 20 s | Elderly 60 s | Child 45 s

Scenario	Standard	Elderly	Child	Why this configuration?
Day / transition + no motion	• Blinds follow daylight priority – Dark room: OPEN – Medium room: HALF – Bright room: CLOSED • Automatic ambient light: OFF	• Blinds follow daylight priority – Dark room: OPEN – Medium room: HALF – Bright room: CLOSED • Automatic ambient light: OFF	• Blinds follow daylight priority – Dark room: OPEN – Medium room: HALF – Bright room: CLOSED • Automatic ambient light: OFF	Blinds maximise useful daylight during the day. With no motion, the system keeps ambient light OFF to save energy and maintain natural comfort.
Day / transition + motion detected	• Blinds follow daylight priority – Dark room: OPEN – Medium room: HALF – Bright room: CLOSED • If room is dark: ambient light gives DIM support • If room is medium/bright: no ambient light	• Blinds follow daylight priority – Dark room: OPEN – Medium room: HALF – Bright room: CLOSED • If room is dark: ambient light gives DIM support • If room is medium/bright: no ambient light	• Blinds follow daylight priority – Dark room: OPEN – Medium room: HALF – Bright room: CLOSED • If room is dark: ambient light gives DIM support • If room is medium/bright: no ambient light	Daylight is prioritised. When the room is still dark and someone is present, DIM ambient light supports visibility while preserving daylight and comfort.
Night + no motion	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	At night, blinds stay closed for privacy and rest. With no motion, ambient light remains OFF to avoid unnecessary light and save energy.
Night + motion detected	• Blinds: CLOSED • If room is dark: ambient light ON • If room is medium: ambient light DIM	• Blinds: CLOSED • If room is dark: ambient light DIM • If room is medium: ambient light DIM	• Blinds: CLOSED • If room is dark: ambient light DIM • If room is medium: ambient light DIM	Night-time visibility is supported based on user profile. Elderly and child profiles use gentler light to reduce glare and improve comfort and safety.

Manual lamp override: if the manual lamp is ON, the automatic ambient light stays OFF.

Figure 7: Bedroom Normal mode decision table for lighting and blinds.



SLEEP Mode - Behaviour by Profile

Decision table for the smart bedroom in SLEEP mode

Purpose: maximise rest and minimise disturbance

Presence hold: Standard 20 s | Elderly 60 s | Child 45 s

Scenario	Standard	Elderly	Child	Why this configuration?
Day / transition + no motion	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	Sleep mode prioritises rest during the day/transition period. With no motion, the room remains quiet and dark.
Day / transition + motion detected	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	Even if motion is detected, sleep mode suppresses daytime lighting support so rest remains the main priority.
Night + no motion	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • Automatic ambient light: OFF	At night with no motion, the room stays fully dark to maximise rest and avoid unnecessary light.
Night + motion detected	• Blinds: CLOSED • Automatic ambient light: OFF	• Blinds: CLOSED • If room is dark: ambient light DIM	• Blinds: CLOSED • If room is dark: ambient light DIM	Vulnerable users receive DIM support lighting at night to improve safety and reassurance, while Standard profile keeps lighting OFF to minimise disturbance.

Note: Sleep mode gives maximum priority to rest. Vulnerable users may still receive DIM night support for safety.

Figure 8: Bedroom Sleep mode decision table for lighting and blinds.

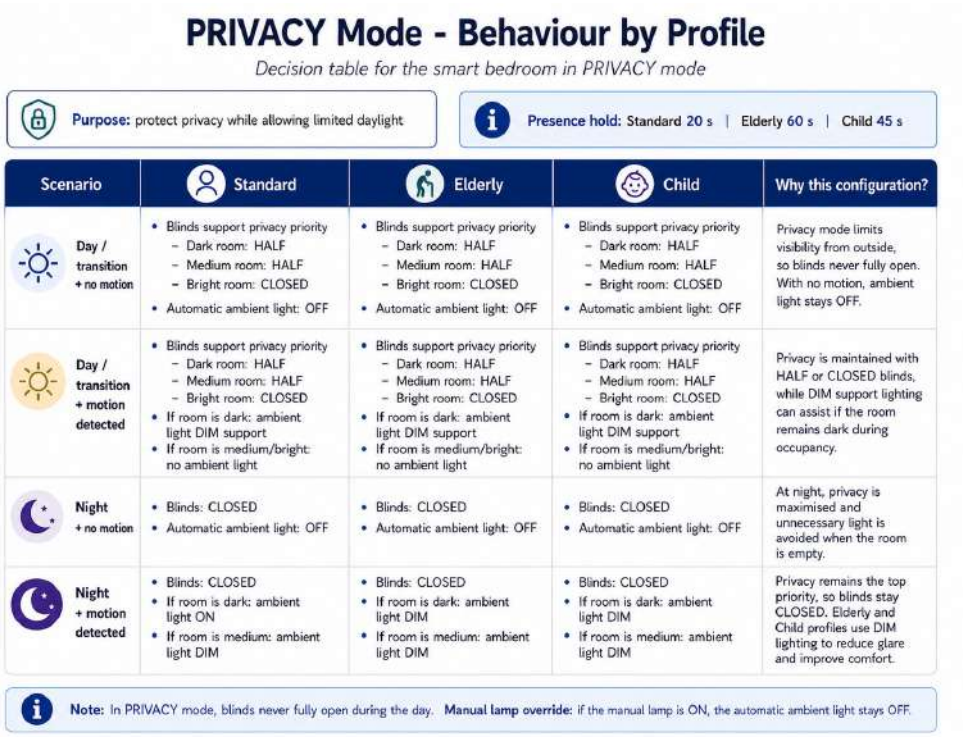


Figure 9: Bedroom Privacy mode decision table for lighting and blinds. In Privacy mode, half-open blinds allow natural light to enter the room while still preventing visibility from outside. This ensures that indoor illumination can be maintained without compromising privacy at any stage.

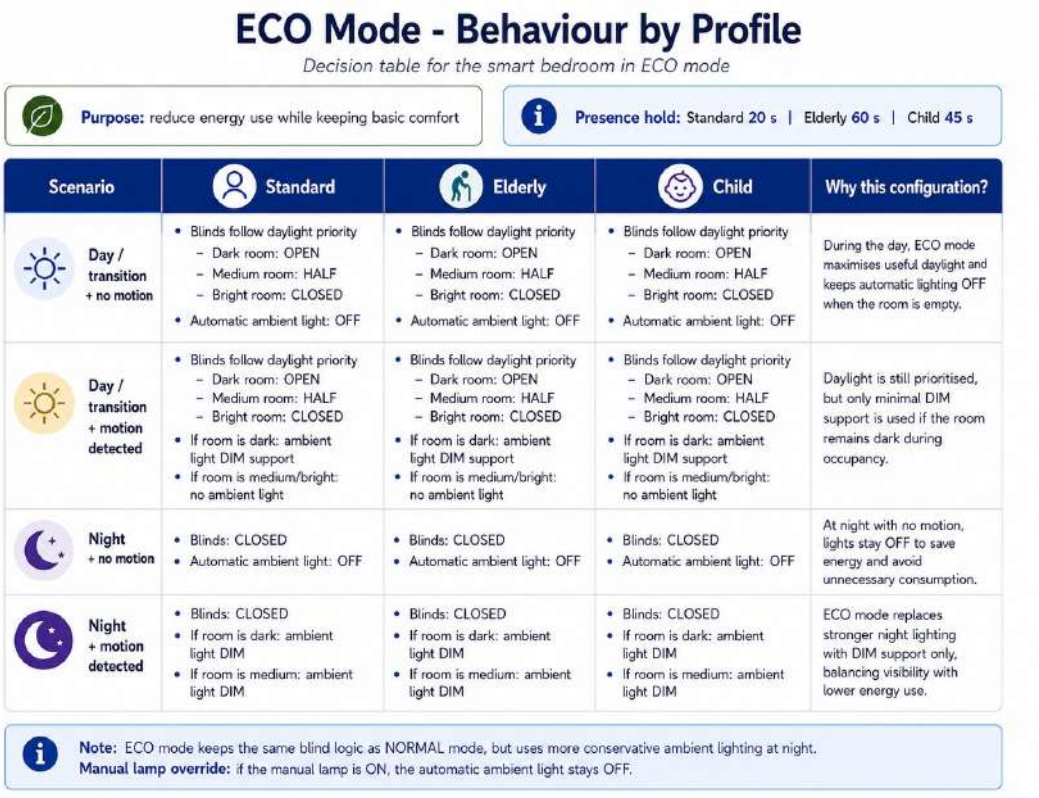


Figure 10: Bedroom Eco mode decision table for lighting and blinds.

3.4.5 Bedroom Thermal Logic

Bedroom thermal comfort is managed through DHT22 sensing and AC relay control. The response varies by profile and mode: vulnerable user profiles justify slightly earlier intervention, while *Eco* delays activation to reduce energy use. The following figures summarise the designed AC behaviour.

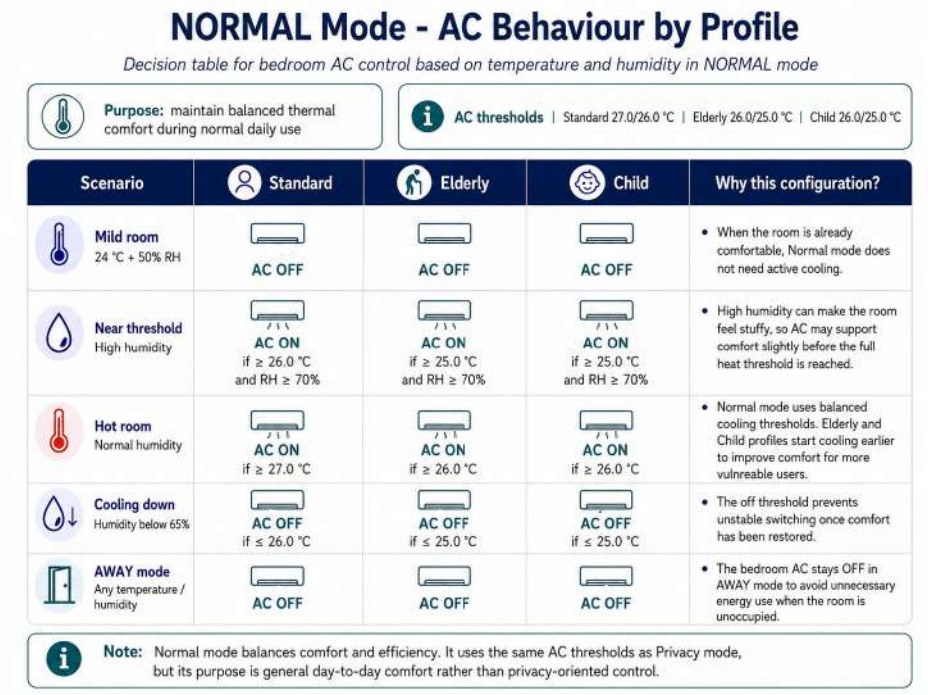


Figure 11: Bedroom Normal mode AC behaviour by profile.

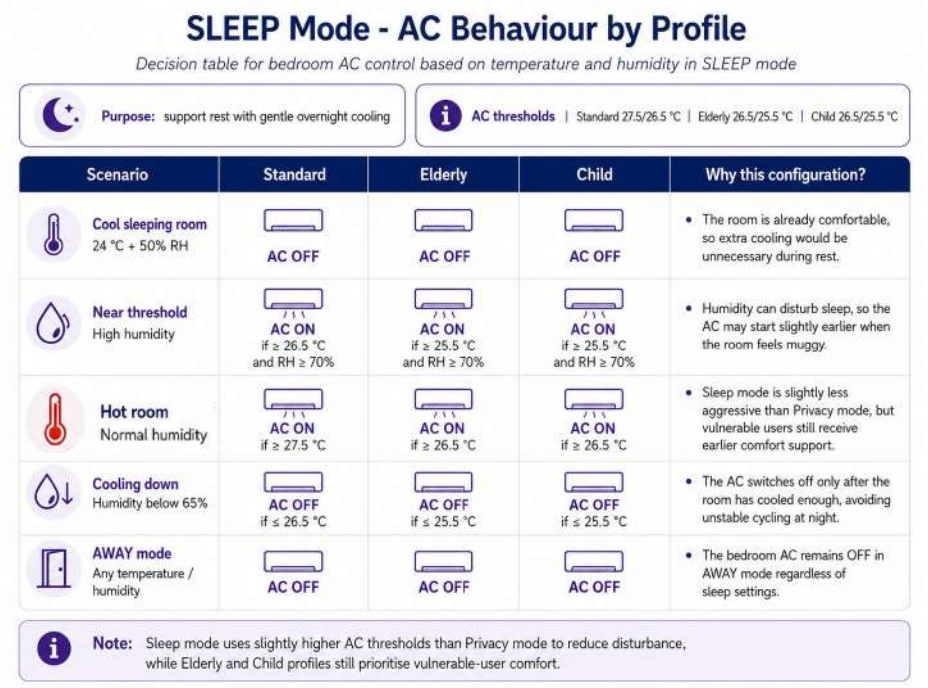


Figure 12: Bedroom Sleep mode AC behaviour by profile.

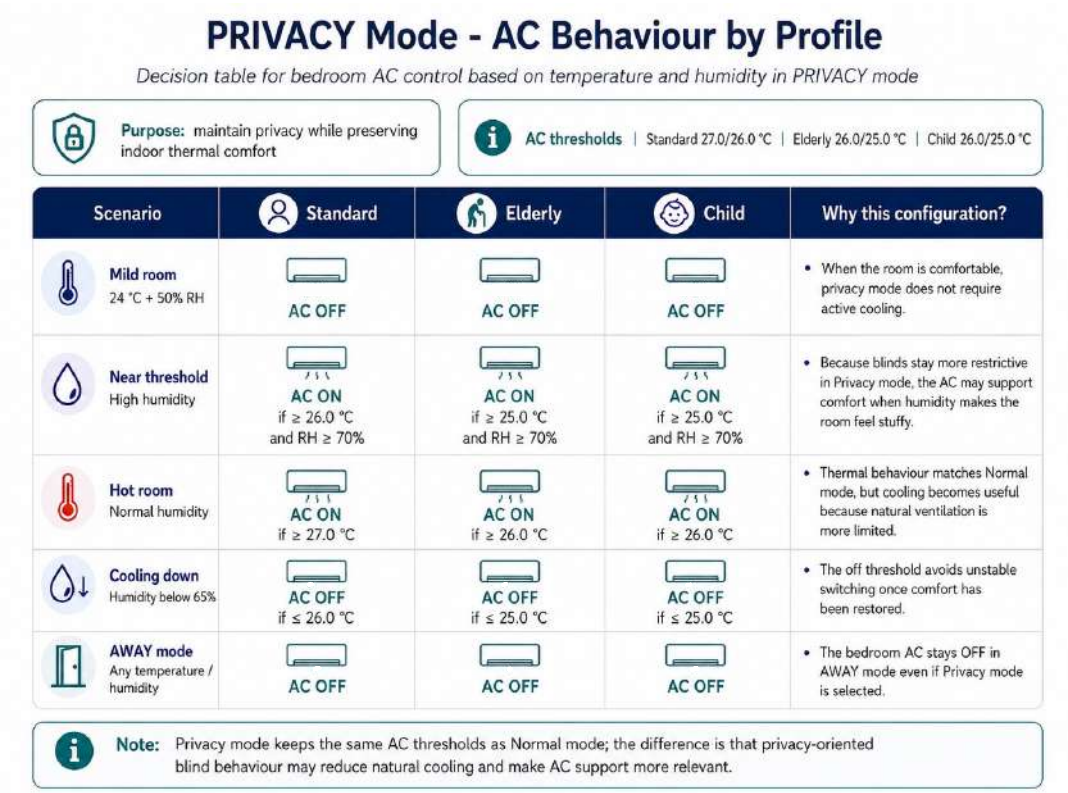


Figure 13: Bedroom Privacy mode AC behaviour by profile.

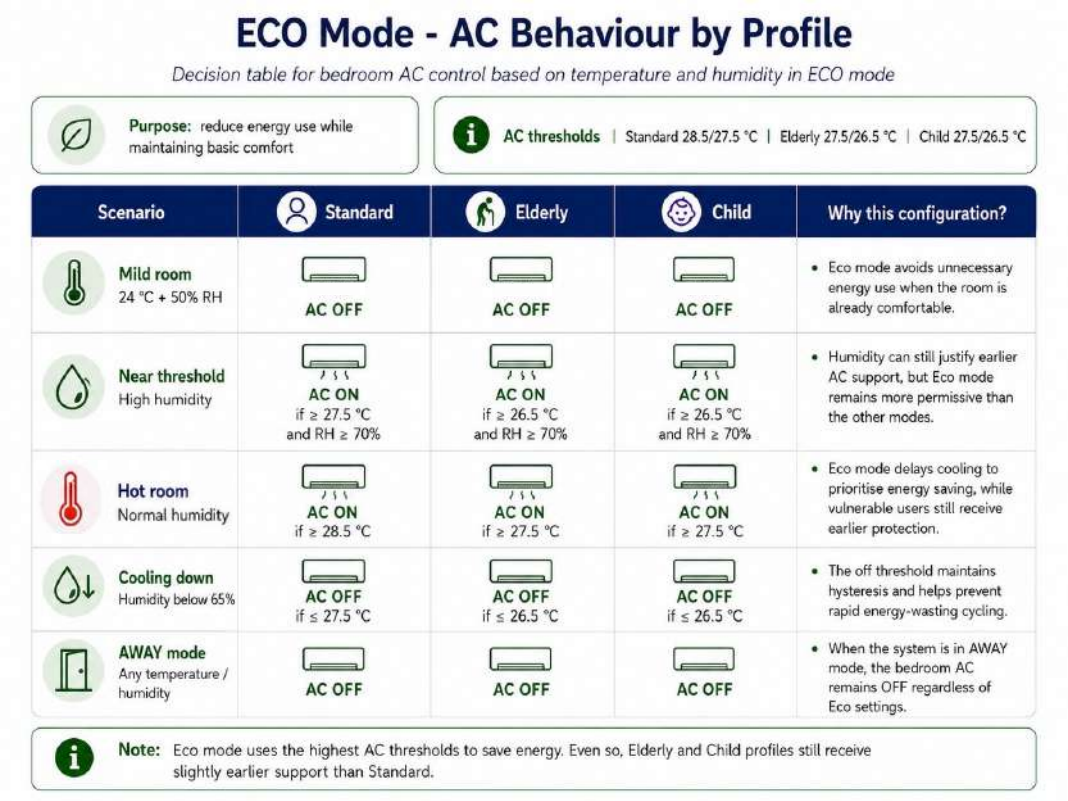


Figure 14: Bedroom Eco mode AC behaviour by profile.

3.5 Kitchen / Living Room Module

3.5.1 Design Rationale

The kitchen/living room follows the same general philosophy as the bedroom but serves a different role. It is a shared, active, and safety-sensitive environment. For this reason, the design avoids bedroom-style automatic ambient support lighting and instead uses a single manually controlled main lamp. The main priorities are functional control, thermal comfort, occupancy awareness, gas safety monitoring, and emergency response.

This module is also the strongest candidate for machine-learning integration because it combines several environmental signals that can describe risk together. Unlike the bedroom, where most behaviour can be explained through comfort and privacy rules, the kitchen/living room benefits from a hybrid design that combines rule-based safety logic with an Edge Impulse classifier.

3.5.2 Hardware Structure

The kitchen/living room module includes an ESP32-S3, PIR motion sensor, indoor and outdoor LDR sensors, MQ2 gas sensor (Zhengzhou Winsen Electronics Technology Co., Ltd., n.d.), DHT22 temperature and humidity sensor (Aosong Electronics, n.d.), servo-controlled blinds, manual main lamp output, AC relay, buzzer, LCD display, sprinkler or emergency-cooling indicator LED, and buttons for mode, profile, house state, and child mode interaction.

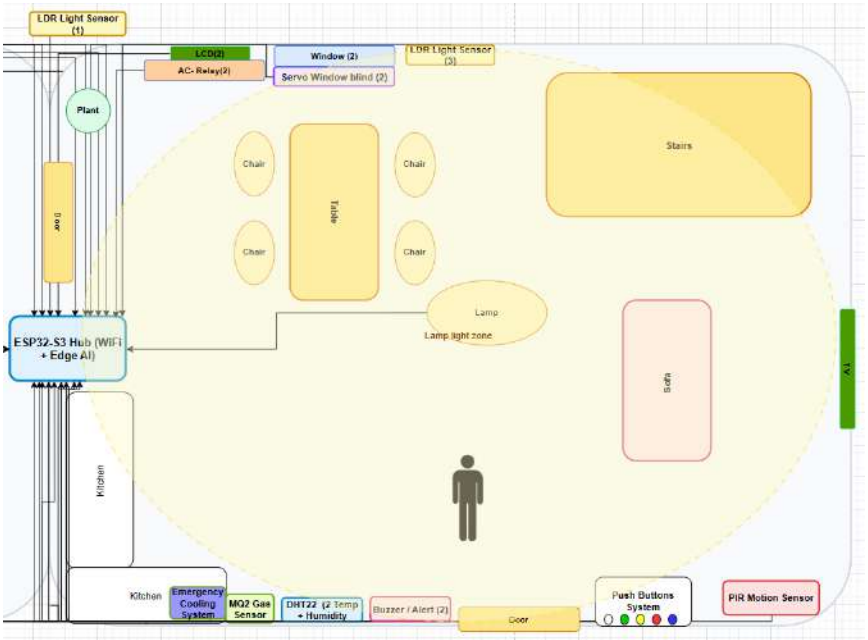


Figure 15: Kitchen/living room hardware layout and logical arrangement in draw.io.



(a) 2D plan view.

(b) 3D view 1.

(c) 3D view 2.

(d) 3D view 3.

Figure 16: Kitchen/living room visualisation created in Planner 5D, including the 2D plan and three dimensional views. Available at: Planner 5D.

3.5.3 Kitchen/Living Behavioural Logic

The final kitchen/living room design uses three room modes: *Normal*, *Privacy*, and *Eco*. *Sleep* was removed because it added little value in a shared functional room. The module still keeps the same profile logic as the rest of the house: *Standard* gives the default response, while *Elderly* and *Child* apply slightly more cautious thresholds and longer response timings. The room also supports the global *Home/Away* state, so repeated motion while armed can escalate from suspicious motion to intruder alert.

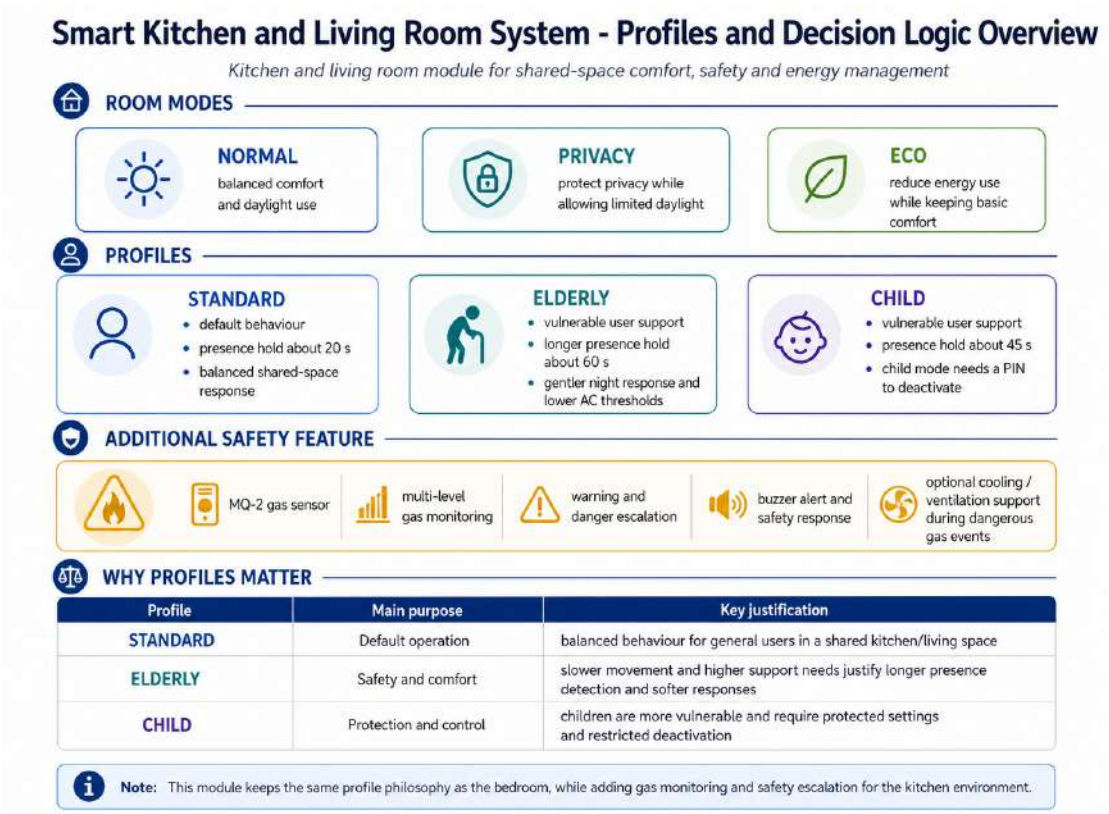


Figure 17: Kitchen/living room system overview. This shared space module follows the same overall house logic used in the bedroom, including the *Normal*, *Privacy*, and *Eco* modes, together with the *Standard*, *Elderly*, and *Child* profiles. However, *Sleep* mode is not implemented in this area, as the kitchen/living room is not mainly intended for night time rest. Unlike the bedroom, this module does not use automatic ambient lighting and relies on a manually controlled main lamp. Alongside blind and thermal control, the module integrates gas monitoring, heat anomaly detection, and emergency cooling response.

3.5.4 Gas and Environmental Safety Logic

The kitchen/living room adds the main safety dimension of the house: gas and environmental risk detection. Because a low-cost MQ2 sensor cannot cleanly separate harmless vapour, cooking fumes, and dangerous events, the system does not depend on one instant reading. This interpretation is consistent with using the MQ2 as a broad combustible-gas sensing element rather than as a precise domestic gas analyser (Zhengzhou Winsen Electronics Technology Co., Ltd., n.d.). Instead, the original rule-based logic uses thresholds, persistence timers, and staged escalation. The final design extends this with an Edge Impulse classifier using three input features:

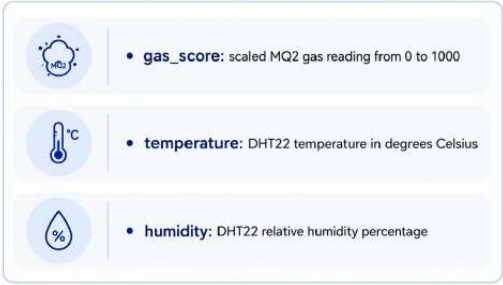


Figure 18: Input features.

The model predicts five environmental safety states:

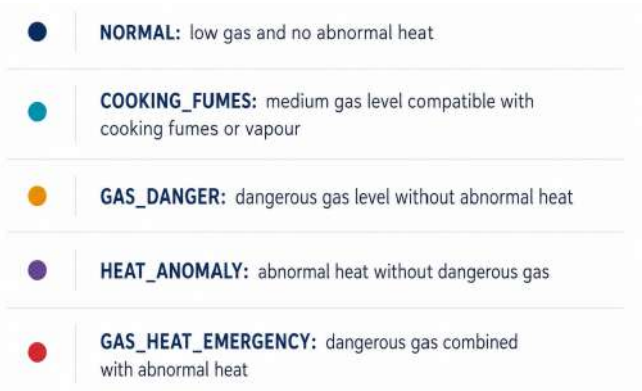


Figure 19: Output classes.

This gives temperature a clearer role in the logic: abnormal heat without dangerous gas becomes `HEAT_ANOMALY`, while dangerous gas combined with abnormal heat becomes `GAS_HEAT_EMERGENCY`. The latter activates the emergency-cooling response represented by a blinking sprinkler LED. The result is a hybrid safety structure in which the gas state machine remains the deterministic fallback while the AI layer provides faster contextual interpretation.

3.5.5 Kitchen/Living Thermal Logic

The kitchen/living room also uses DHT22-based thermal monitoring and AC control. This logic is simpler than in the bedroom because the main focus is shared-space comfort and safety rather than rest support. *Eco* delays AC intervention to reduce energy use, while the *Elderly* and *Child* profiles justify slightly more cautious thresholds. Safety overrides take priority: when cooking fumes or gas danger are active, the AC is switched off so that potentially contaminated air is not circulated through the room.

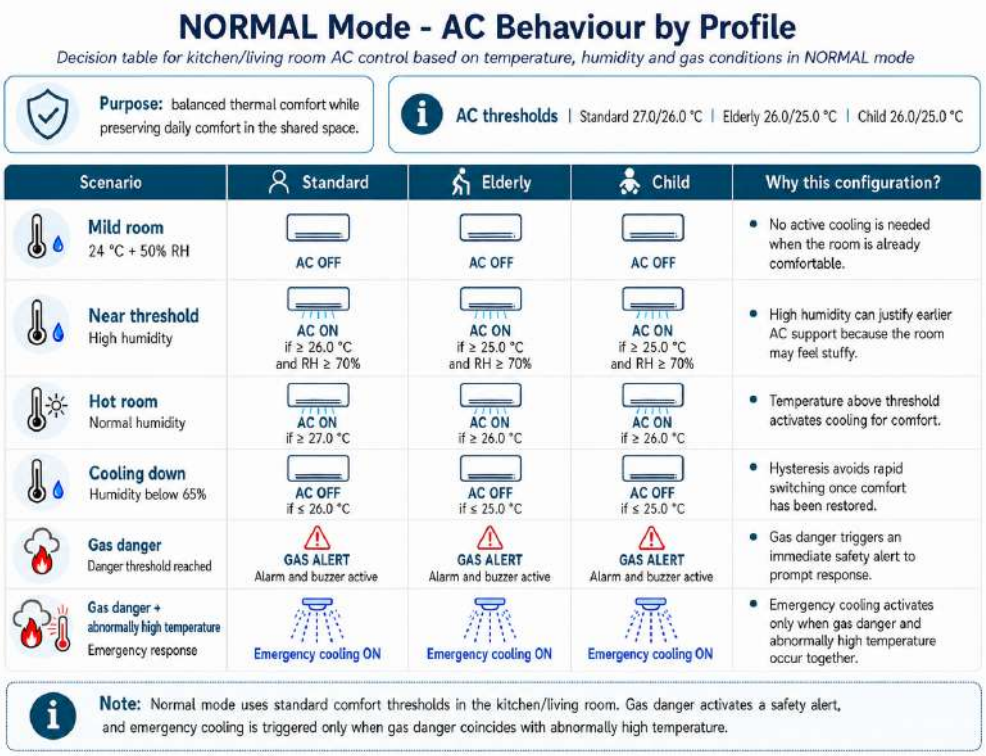


Figure 20: Kitchen/living room Normal mode AC behaviour by profile.

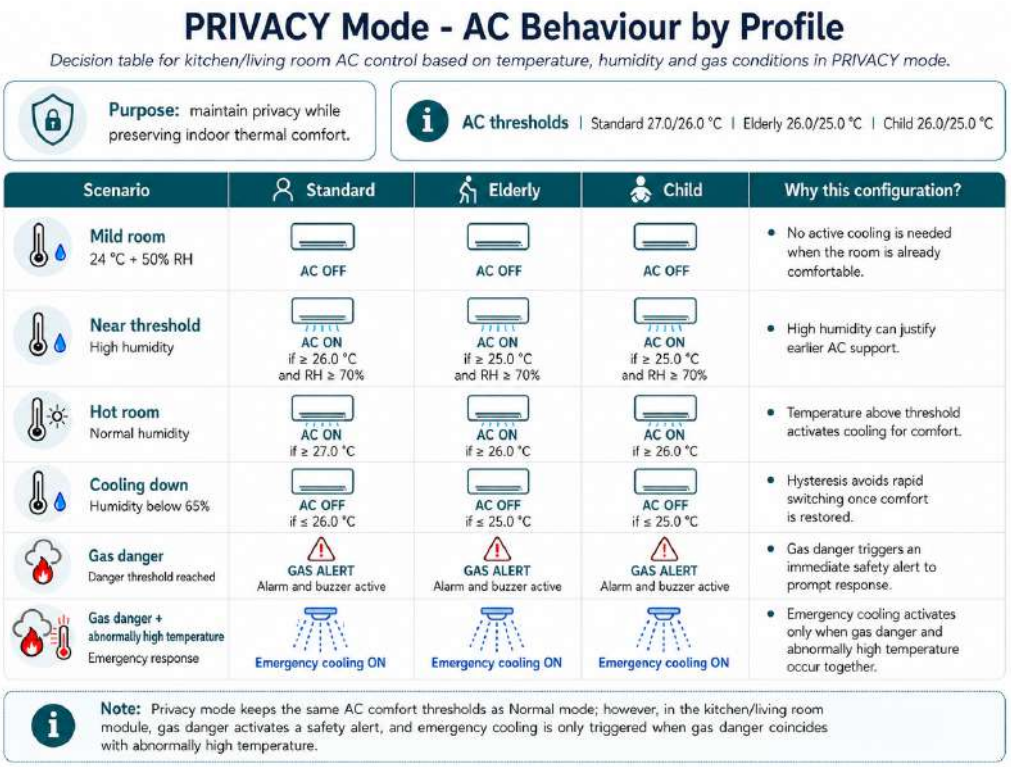


Figure 21: Kitchen/living room Privacy mode AC behaviour by profile.

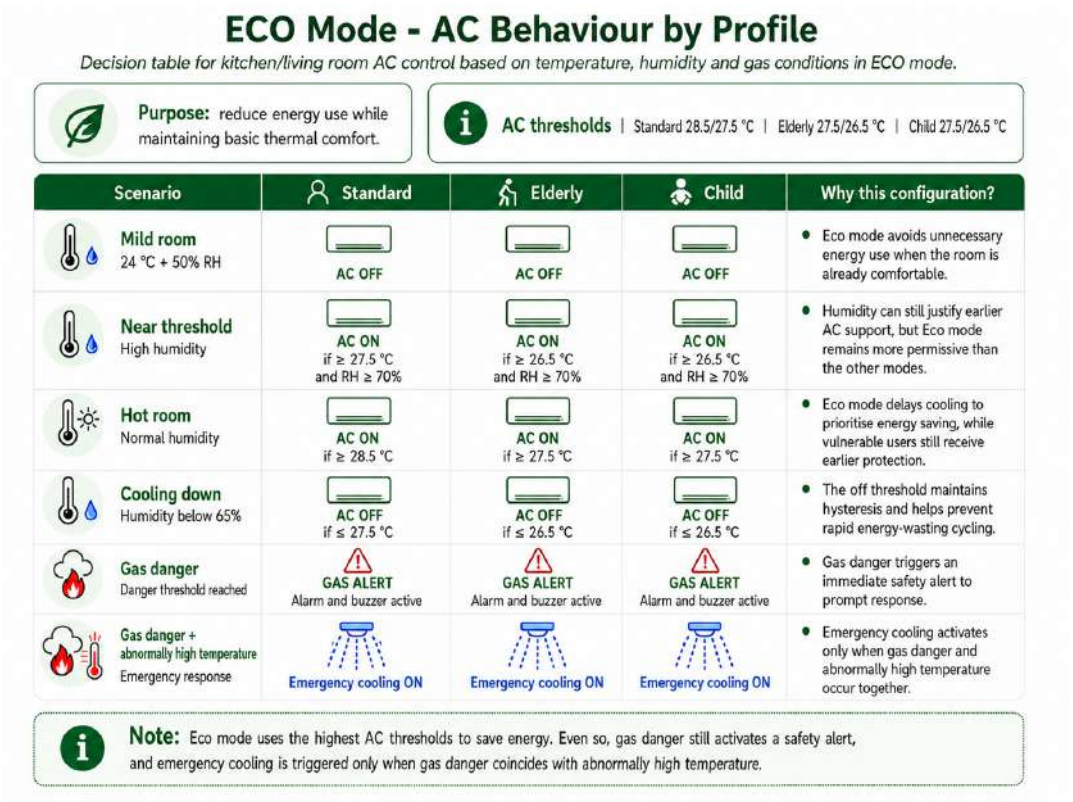


Figure 22: Kitchen/living room Eco mode AC behaviour by profile.

4 AI Model Design & Deployment

The AI part of the project is deliberately limited to the most safety-relevant and data-rich subsystem: environmental safety classification in the kitchen/living room. This keeps the task focused, explainable, and suitable for embedded deployment.

4.1 Role of AI in the Project

The AI component is not intended to replace the rule-based control system. The original rules remain important because they are transparent, deterministic, and appropriate for fallback behaviour in safety-related situations. In the current prototype, Edge Impulse is used to convert the existing environmental logic into a deployable embedded classifier based on `gas_score`, `temperature`, and `humidity` (Edge Impulse, n.d.a). Its role is to add a compact environmental interpretation layer that can be deployed on the ESP32-S3 and connected to the wider room behaviour.

The resulting workflow covers dataset creation, labelling, model training, testing, performance evaluation, and Arduino-library export for ESP32-S3 deployment. Because the dataset in this version is simulated and rule-labelled, the model should be treated as an initial Edge Impulse prototype rather than a certified safety system. The safest interpretation is therefore hybrid: AI supports the existing logic while deterministic rules remain available as fallback.

4.2 Problem Type

The selected problem is a multi-class classification task. The model receives three numerical inputs from the kitchen/living room sensors and predicts the current environmental safety state. As mentioned previously, the input features are:

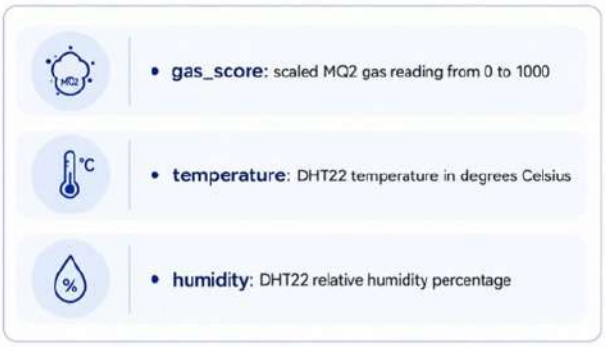


Figure 23: Input features.

The output classes are:

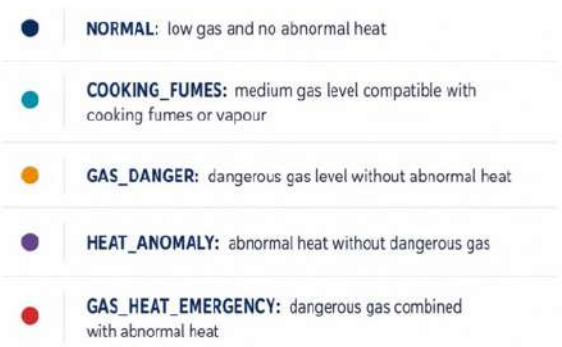


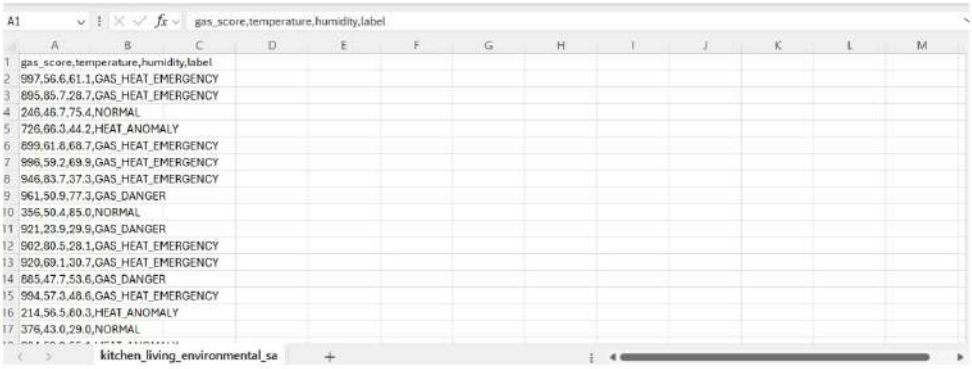
Figure 24: Output classes.

This is more appropriate than a gas-only classifier because it allows the system to distinguish between gas danger, heat anomaly, and a combined gas-and-heat emergency, while also maintaining a direct link between model output and embedded response.

4.3 Dataset Collection and Labelling

A custom dataset was created from the kitchen/living room simulation in Wokwi using the same ESP32-S3, MQ2, and DHT22 setup as the implemented module. This was preferable to a public dataset because the target classes were specific to the environmental safety states defined for this project. Data were collected through a simplified ESP32-S3 sketch that removed Blynk, LCD logic, buttons, and actuator control and printed clean CSV rows in the form:

```
gas_score,temperature,humidity,label
```



	A	B	C	D	E	F	G	H	I	J	K	L	M
1	gas_score,temperature,humidity,label												
2	997,56.6,61.1,GAS_HEAT_EMERGENCY												
3	895,85.7,28.7,GAS_HEAT_EMERGENCY												
4	246,46.7,75.4,NORMAL												
5	726,66.3,44.2,HEAT_ANOMALY												
6	899,61.6,68.7,GAS_HEAT_EMERGENCY												
7	986,59.2,69.9,GAS_HEAT_EMERGENCY												
8	946,83.7,37.3,GAS_HEAT_EMERGENCY												
9	961,50.9,77.3,GAS_DANGER												
10	356,50.4,85.0,NORMAL												
11	921,23.9,29.9,GAS_DANGER												
12	902,80.5,28.1,GAS_HEAT_EMERGENCY												
13	920,69.1,30.7,GAS_HEAT_EMERGENCY												
14	885,47.7,53.6,GAS_DANGER												
15	994,57.3,48.6,GAS_HEAT_EMERGENCY												
16	214,56.5,60.3,HEAT_ANOMALY												
17	376,43.0,29.0,NORMAL												

Figure 25: Dataset table used for the environmental safety classifier.

The simulated gas level, temperature, and humidity were varied to produce examples of the five environmental safety states. Labels were generated automatically from the same safety definitions used in the embedded system, which avoided inconsistent manual annotation and ensured alignment between dataset and designed behaviour.

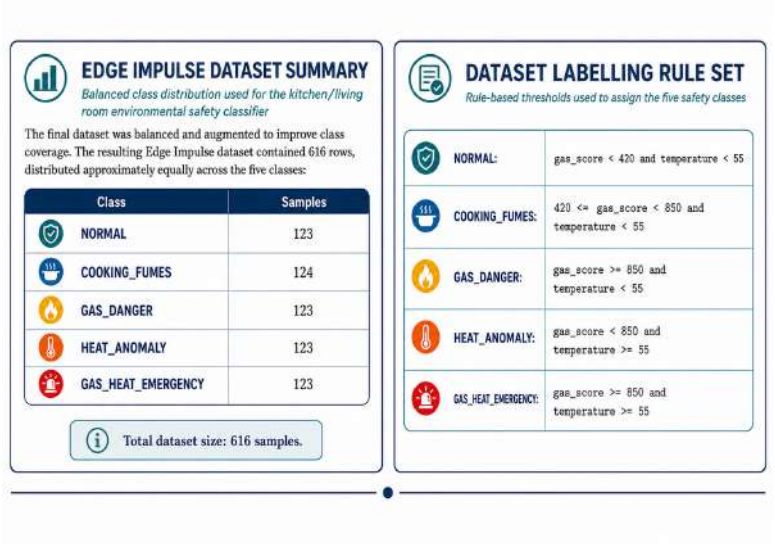


Figure 26: Edge Impulse dataset summary and rule-based labelling criteria for the kitchen/living room environmental safety classifier. The left panel shows the final balanced class distribution across the five target classes, while the right panel presents the threshold based rules used to assign labels from gas score and temperature conditions before model training.

Humidity was retained as a contextual feature even though labels were driven mainly by gas score and temperature.

4.4 Impulse Design

The Edge Impulse pipeline was configured as a small tabular classification impulse using the three numerical features `gas_score`, `temperature`, and `humidity`. The processing stage used raw data features with standard scaling, and feature-importance analysis was used to assess the contribution of each input.

The results matched the intended system logic: gas score was the most important feature, temperature the second most important, and humidity a smaller contextual feature. The learning block used a small neural-network classifier with three input features, two dense layers, and five output classes. This was appropriate because the dataset is small, numerical, and low-dimensional, so a larger model would have added complexity without clear deployment benefit.

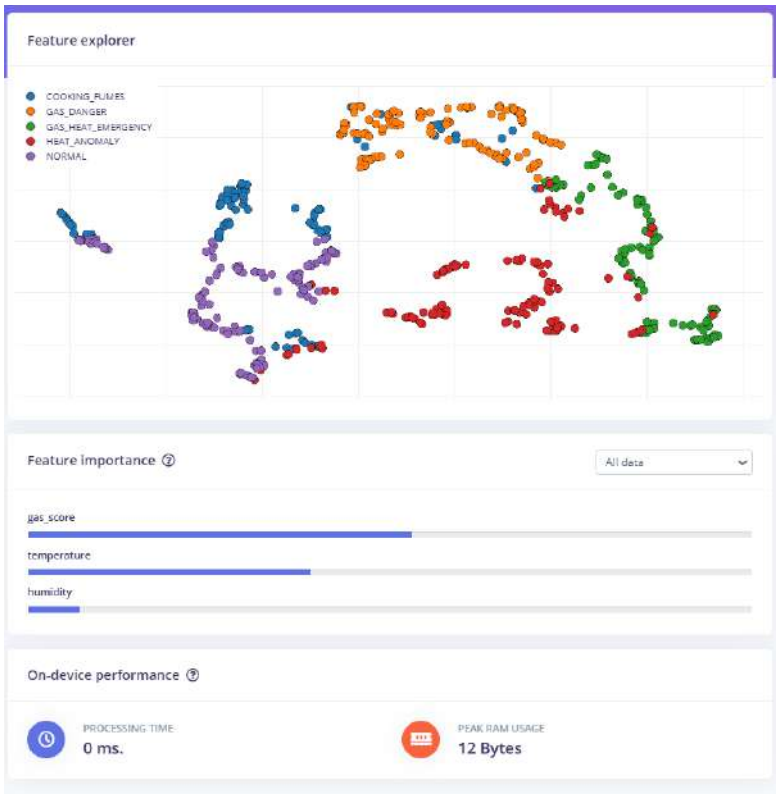


Figure 27: Edge Impulse outputs for the kitchen/living room environmental safety classifier. The upper part shows the feature explorer and the generated feature space for the five environmental safety classes. The lower part shows the feature importance analysis, where `gas_score` is the most influential feature, followed by `temperature` and `humidity`.

4.5 Training and Performance Metrics

The classifier was trained in Edge Impulse using 100 training cycles and a learning rate of 0.0005. Training reached approximately 93.4% accuracy with a weighted F1 score of 0.93, while the held-out test set achieved approximately 85.9% accuracy, an AUC of 0.99, weighted precision of 0.91, weighted recall of 0.89, and weighted F1 score of 0.89. The strongest performance was observed in the most safety-critical classes, especially `NORMAL`, `GAS_DANGER`, and `GAS_HEAT_EMERGENCY`. The main weakness was the intermediate `COOKING_FUMES` class, which is expected because it sits between normal and dangerous gas levels.

Edge Impulse also reported embedded performance estimates suitable for ESP32-S3 deployment: approximately 1 ms inference time, 1.4 KB peak RAM, and 14.4 KB flash usage. These values indicate that the model is lightweight enough for embedded inference.



Figure 28: Edge Impulse training results for the environmental safety classifier. The model achieved approximately 93.4% training accuracy with strong performance on the most safety critical classes.

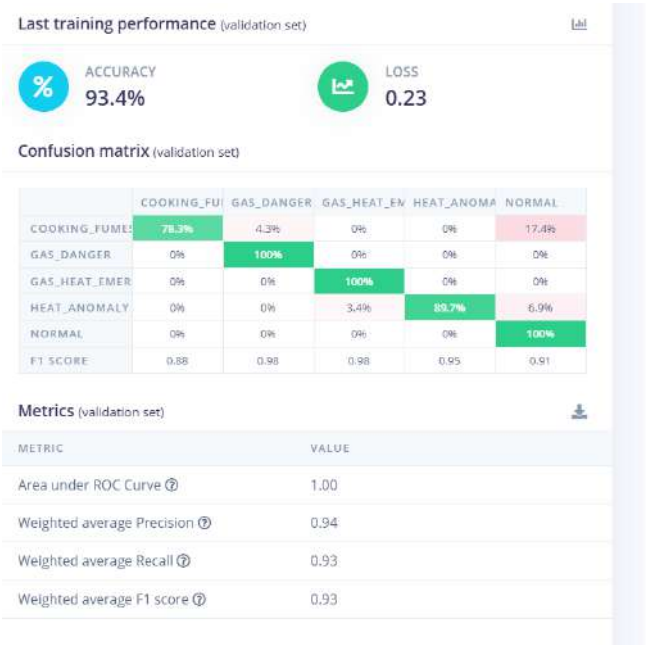


Figure 29: Training confusion matrix for the Edge Impulse classifier. The matrix shows strong classification performance for `NORMAL`, `GAS_DANGER`, and `GAS_HEAT_EMERGENCY`, with most ambiguity occurring around the intermediate `COOKING_FUMES` class.

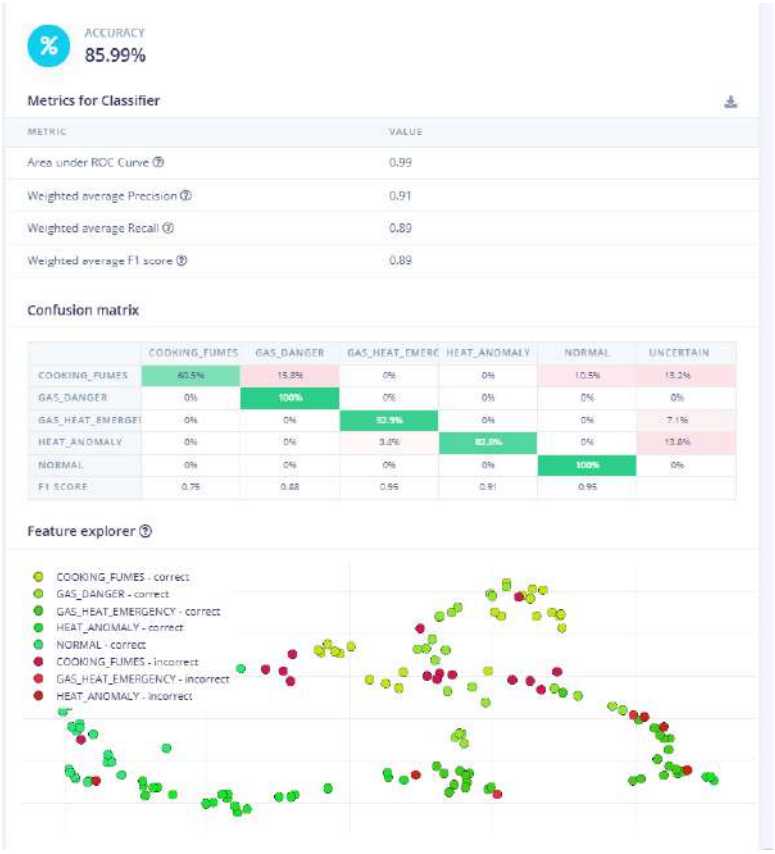


Figure 30: Model testing results on the held out test set. The Edge Impulse model achieved approximately 85.9% test accuracy, with an AUC of 0.99 and strong performance on the main safety critical classes.

4.6 Deployment Strategy

The trained model was exported from Edge Impulse as an Arduino library using quantised int8 optimisation and the EON compiler. The generated library was prepared for ESP32-S3 deployment and integrated into the kitchen/living room code as an additional AI layer. The embedded system uses the model output to update an environmental safety state variable that then informs the LCD, Blynk dashboard, buzzer, blinds, AC logic, and emergency-cooling response (Edge Impulse, n.d.b; Espressif Systems, n.d.).

In Wokwi, the full Edge Impulse runtime caused simulator instability when the generated classifier was executed. For that reason, the simulation version uses a Wokwi-safe inference mode that follows the same five environmental safety classes as the trained model. This allows system-level behaviour to be tested in simulation without crashing the virtual ESP32-S3. On real hardware, the exported Arduino library can be enabled for full inference testing. The final architecture therefore remains hybrid: rule-based safety logic stays active as deterministic fallback while the model provides contextual environmental classification.

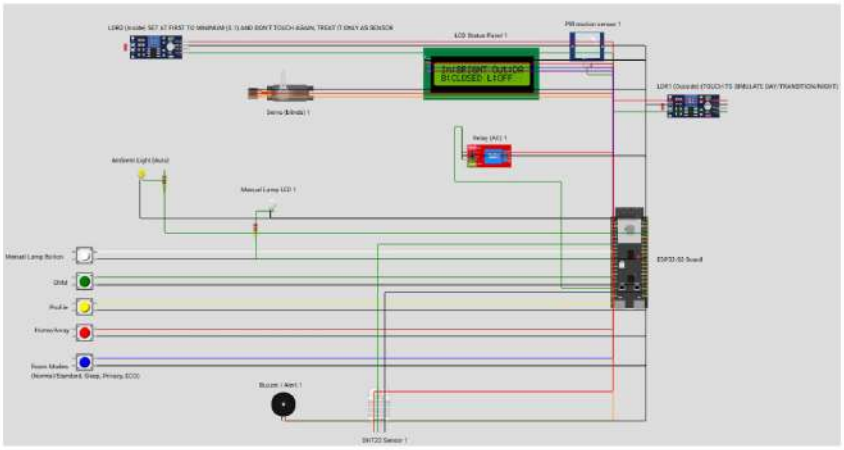
5 Implementation

5.1 Simulation Environment

The practical implementation was developed mainly in Wokwi. This allowed sensor integration, room logic, LCD outputs, serial monitoring, Blynk connectivity, and safety-related behaviour to be tested without full physical hardware assembly. Wokwi was especially useful for iterative debugging, threshold tuning, and controlled testing of gas, temperature, humidity, light, and motion changes (Wokwi, n.d.).

5.2 Bedroom Module Implementation

The bedroom module was the most behaviourally detailed subsystem developed during the earlier stage of the project. It implements light-driven state classification, PIR-based occupancy detection with profile-dependent hold times, adaptive blinds, ambient support lighting, manual lamp integration, AC control through DHT22 readings, and local feedback through the LCD and Serial Monitor. It also supports child-mode protection, Away-mode behaviour, and authenticated state changes.



(a) Wokwi components and circuit connections.

```
======== SYSTEM STATUS (refresh every 5 seconds) =========
Room mode: NORMAL
Profile: STANDARD
House status: HOME
Pet at home: NO
Alert: NONE
AWAY motion count: 0
LDR2 inside raw: 2205 | inside effective brightness score: 711
LDR1 outside raw: 4063 | active outside score: 0 | source: BLYNK_V3
Inside state: BRIGHT
Outside state: NIGHT
PIR raw: 0
Presence active: YES
Manual lamp: ON
Blinds: CLOSED
Smart ambient light: OFF
Temperature: 60.7 C
Humidity: 100.0 %
Thermal state: WARM
AC: ON
Request active: NONE
```

(b) Serial Monitor outputs.

Figure 31: Bedroom implementation in Wokwi, showing the components and the Serial Monitor outputs.

5.3 Kitchen/Living Room Implementation

The kitchen/living room implementation follows the same broad software structure but adapts it to different priorities. Its main additions are gas sensing, emergency safety escalation, and Edge Impulse-based environmental classification. The room keeps a manual main lamp instead of automated ambient support lighting and combines rule-based thresholds with AI-derived environmental state interpretation.

The implementation includes DHT22 thermal sensing, MQ2 gas-score conversion, PIR motion detection, LDR-based brightness interpretation, servo blind control, AC relay behaviour, buzzer alerts, sprinkler or emergency-cooling LED indication, LCD output, Blynk monitoring, and the same Home/Away security logic used elsewhere in the house. The environmental-state pipeline is summarised below.

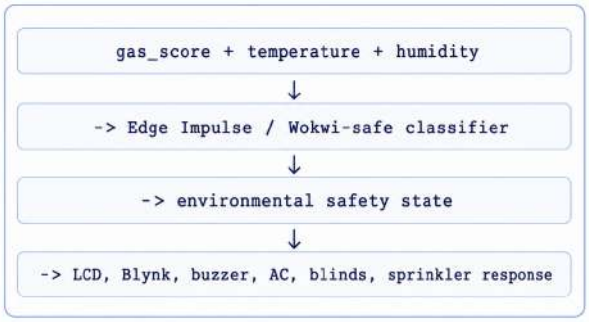
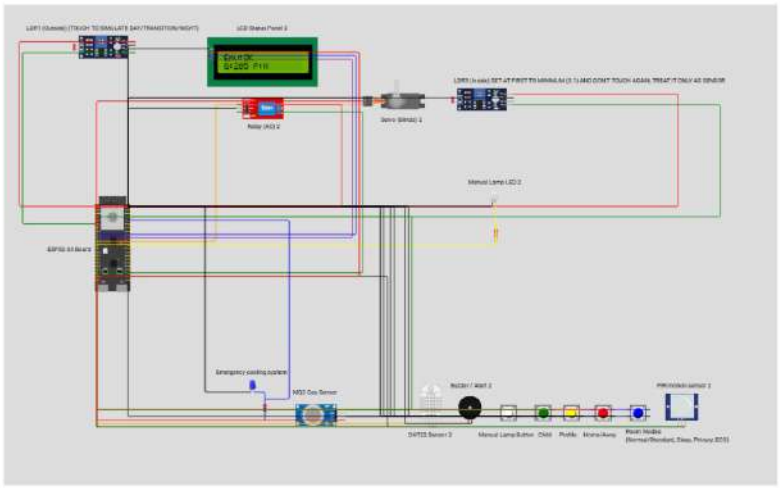


Figure 32: Environmental state pipeline used for the kitchen/living room AI safety layer.

Simulation confirmed the intended hybrid behaviour. For example, when gas score rose to around 885 and temperature to 72.4 °C, the system entered `GAS_HEAT_EMERGENCY`, activated the emergency-cooling state, opened the blinds, turned off the AC, and displayed the alert as *GAS + HIGH TEMP*. When gas later decreased while temperature remained high, the state moved to `HEAT_ANOMALY`, and when both returned to safe values, the status returned to `NORMAL`.



(a) Wokwi components and circuit connections.

```
-------- KITCHEN/LIVING STATUS (refresh every 5 seconds) ---------
Room mode: NORMAL
Profile: STANDARD
House status: HOME
Pet at home: NO
Alert: NONE
AWAY motion count: 0
LDR inside raw: 4063 | inside effective brightness score: 432
LDR outside raw: 610 | active outside score: 851 | source: WOKWI_LDR
Inside state: BRIGHT
Outside state: DAY
PIR raw: 0
Presence active: NO
Manual main lamp: OFF
Blinds: OPEN
MQ2 raw: 843 | gas score: 205
MQ2 warming up: NO
Gas thresholds - clear/warn/danger/severe: 350 / 420 / 850 / 970
Gas status: NORMAL
AI/environment status: NORMAL
Last ML prediction: NORMAL
ML valid: YES
ML confidence: 0.990
Emergency cooling threshold: 55.0 C
Emergency cooling active: NO
Temperature: 18.3 C
Humidity: 40.0 %
Thermal state: COOL
AC: OFF
Request active: NONE
AI environmental status (Wokwi-safe): NORMAL confidence=0.990
```

(b) Serial Monitor outputs.

Figure 33: Kitchen/living room implementation in Wokwi, showing the components and the Serial Monitor outputs.

5.4 Blynk Remote Monitoring Layer

Blynk was used as the remote monitoring and interaction layer for the two ESP32-S3 room modules. The main decision logic remains local on the microcontroller, while Blynk provides a dashboard for observing sensor values, interpreted room states, alerts and selected user controls from a web or mobile interface.

This separation is important for the overall design: Blynk improves visibility and interaction, but it does not replace the embedded room logic. Comfort decisions, safety escalation, alarm behaviour and protected state transitions are still handled by the local ESP32-S3 code, so the system remains understandable and partially resilient even if the dashboard layer is unavailable.

Dashboard role in each module

- **Bedroom module:** Blynk is used to monitor temperature, humidity, brightness-related values, PIR-based occupancy, room mode, active profile, house status, blind state, smart-light state, AC state, thermal state, alerts, pet status and Away-mode motion count.
- **Kitchen and living room module:** Blynk is used to monitor temperature, humidity, gas score, gas-related safety state, environmental safety classification, emergency-cooling state, manual lamp state, AC state, alerts and selected control variables.
- **Interaction layer:** dashboard controls can change selected modes or request protected transitions, while sensitive actions such as Child-mode exit and Away-to-Home recovery remain protected through PIN or command-entry logic.

Public repository sanitisation

The public version intentionally removes real Blynk authentication tokens, private credentials, dashboard screenshots and detailed virtual-pin mapping figures. The source code keeps placeholder credentials and clear comments so that another developer can reproduce the structure using their own Blynk template and token.

Virtual pins are treated as implementation details rather than secrets. They remain visible where needed in the source code, but the public report avoids presenting a full dashboard map because it adds operational detail without improving the main technical explanation. A separate repository document can be used if a concise virtual-pin reference is required.

Safe public documentation approach

Item	Public version approach
Blynk tokens	Removed and replaced with placeholders
Dashboard screenshots	Removed from the public report
Virtual-pin mapping	Kept out of the report; documented only at a high level
Main value shown	Remote monitoring, user interaction and transparency of room states

Note: this public report describes Blynk at a system level. The GitHub repository contains the sanitised sketches and README files needed to understand or adapt the dashboard integration without exposing private credentials.

Dashboard screenshots and detailed virtual-pin mapping figures have been removed from this public version to avoid publishing unnecessary interface details.

5.5 Authentication and Security Logic

A recurring feature across the implementation is local authentication for sensitive state changes. Child-mode deactivation and Away-to-Home transition require authentication through the serial interface or Blynk command input, while intruder-alert states use a separate secret-word mechanism for controlled clearance. This adds a modest but meaningful security layer and reinforces the idea that some room states should not be reversible through accidental input.

6 Evaluation

The evaluation is based on scenario testing, simulation behaviour, and Edge Impulse metrics. The aim is not to present a finished commercial product but to assess whether each room behaves coherently under changing environmental and user conditions, and whether the kitchen/living room AI component produces useful environmental classifications.

In the bedroom, the evaluation focused on night-time behaviour, profile-sensitive support logic, occupancy timing, blind transitions, thermal response, and Away-mode alert escalation. Results show that the same room can respond differently under *Standard*, *Elderly*, and *Child* profiles while maintaining a coherent overall behaviour.

In the kitchen/living room, the evaluation focused on gas-reading behaviour, persistence-based safety escalation, AC interaction, emergency-cooling activation, and the relationship between rule-based logic and AI-derived environmental classification. Simulation confirmed the intended behaviour: high gas and temperature produced `GAS_HEAT_EMERGENCY`, activated the *GAS + HIGH TEMP* alert, opened the blinds, turned off the AC, and triggered emergency cooling; when gas fell but temperature remained high, the state moved to `HEAT_ANOMALY`; when both returned to safe values, the system returned to `NORMAL`.

The Edge Impulse model achieved approximately 93.4% training accuracy and 85.9% test accuracy. The held-out test set produced an AUC of 0.99, weighted precision of 0.91, weighted recall of 0.89, and weighted F1 score of 0.89. The strongest results appeared in the most safety-critical classes, especially `GAS_DANGER`, `GAS_HEAT_EMERGENCY`, and `NORMAL`, while most remaining errors occurred around `COOKING_FUMES`. Estimated embedded performance was also suitable for ESP32-S3 deployment, at approximately 1 ms inference time, 1.4 KB peak RAM, and 14.4 KB flash usage.

A key strength of the prototype is that it demonstrates differentiated room logic within a shared architectural framework while also implementing an initial Edge Impulse workflow for environmental safety classification. The main limitation is that the dataset is simulated and rule-labelled rather than collected from physical sensor deployments, so real-world validation would still be required before treating the model as a safety-certified system.

7 Ethical, Privacy & Security Considerations

A main motivation for the project is privacy-aware local decision making. By keeping the main room logic on the embedded controller, the system avoids unnecessary transmission of household behavioural data to external services. This does not remove all privacy concerns, especially because Blynk introduces a cloud-connected dashboard layer, but it reduces the default exposure of raw indoor activity data. This follows a GDPR-style data-minimisation approach: only the information needed for room-state interpretation is processed or displayed, and the main decision logic remains local rather than continuously exporting raw household activity data (Information Commissioner's Office, 2023).

Security is addressed at a basic but meaningful level through protected state transitions, Away-mode alert escalation, local alarm mechanisms, and room-level modularity. The modular structure also reduces dependence on one central controller for the whole house.

Ethically, the project recognises the limits of low-cost sensors and small machine-learning models. This is especially important for gas and safety-related sensing, so the system interprets gas readings carefully through staged escalation rather than claiming chemically precise identification. In the same way, the Edge Impulse classifier is treated as a supportive environmental classifier rather than a certified safety device.

The hybrid architecture is ethically stronger than a fully opaque AI-only approach. Rule-based fallbacks remain active for safety-critical behaviour, while the AI model adds contextual classification. If the learned model is unavailable, uncertain, or unstable in a specific environment, deterministic safety logic still remains.

8 Discussion

This project shows that a smart home can be designed as a set of room-level embedded systems rather than as one undifferentiated automation layer. This is useful conceptually because each room can express its own priorities, and practically because each module can be implemented, tested, and refined separately.

The contrast between the bedroom and the kitchen/living room is one of the main strengths of the design. The bedroom demonstrates profile-sensitive behaviour around timing, support lighting, rest, and privacy, while the kitchen/living room is shaped more strongly by functional comfort, environmental safety, and emergency response.

The Edge Impulse workflow also clarifies the role of AI in the project. The model is not used as a decorative addition or as a replacement for all rules. Instead, it is applied to a specific problem where multi-sensor classification is useful: environmental safety interpretation in the kitchen/living room. Because the dataset was created in simulation and labelled using the project's own safety definitions, the model mainly learns the environmental boundaries designed during development. Its value is therefore not that it outperforms the original rules, but that it demonstrates a complete embedded machine-learning workflow and prepares the system for future

real-sensor data. The resulting architecture is hybrid, combining explainable rule-based logic with lightweight learned inference.

At the same time, the prototype remains a staged implementation rather than a finished commercial product. Gas logic still requires real-world calibration, the dataset is simulated and rule-labelled, and the full Edge Impulse runtime could not be executed reliably inside Wokwi. To address this, the Wokwi version uses a simulator-safe inference mode that follows the same five environmental safety classes, while the trained model remains available as an Arduino library for future ESP32-S3 hardware testing.

9 Conclusion & Future Work

This report has presented a modular AIoT smart-home system based on ESP32-S3 room controllers. The project has shown how comfort, privacy, safety, and security can be treated as room-specific behaviours shaped by environment, user context, local control logic, remote monitoring, and lightweight embedded AI.

The bedroom and kitchen/living room modules show that a shared house-wide philosophy can coexist with different room priorities. The bedroom focuses on comfort, privacy, rest, and support for vulnerable users, while the kitchen/living room focuses on shared-space functionality, gas monitoring, heat anomaly detection, and emergency response.

A significant outcome is the Edge Impulse environmental safety classifier for the kitchen/living room. The model classifies five states using gas score, temperature, and humidity, achieves 85.9% test accuracy, and shows suitable embedded performance. The system incorporates this classifier within a hybrid architecture in which rule-based safety logic remains active as fallback.

Future work should focus on five areas: real ESP32-S3 hardware testing with the exported Edge Impulse Arduino library; expansion of the dataset with real MQ2 and DHT22 readings; calibration of gas thresholds and persistence timings using physical hardware; extension of the model with `COLD_ANOMALY` detection and heater response; and further refinement of the Blynk dashboards to improve user feedback, alert history, remote authentication flow, and visual presentation of the already implemented AI-related outputs.

The broader value of the project lies in its architecture. Rather than treating AI as a decorative add-on to conventional automation, the project develops an edge-oriented smart home in which local reasoning is already central, making embedded AI both technically meaningful and conceptually justified.

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